Unit 12: Short-Run Production

1. **One Variable and One Fixed Input: Total Product, Marginal Product of Labor, Average Product of Labor,**
2. **Graphing the Product Curves**
3. **Law of Diminishing Marginal Returns**

Objectives of the Study

After studying this unit, the student will

-acquire a thorough knowledge about short run production including graphical representation

-familiar with law of diminishing marginal returns

Production theory forms the backbone of microeconomic analysis. It explains how firms transform inputs into outputs and how output responds to changes in inputs. In every economy, whether developed or developing, production is essential because it creates goods and services that satisfy human wants. Since resources are scarce and human wants are unlimited, efficient production becomes crucial. Firms must make important decisions regarding how much to produce, how many workers to hire, and how to allocate resources efficiently. These decisions depend largely on understanding the relationship between inputs and outputs. This relationship is studied under the theory of production. One of the most important frameworks in production theory is short-run production, where at least one factor of production is fixed.

Production refers to the process of converting inputs (factors of production) into outputs (goods and services). It is not limited to manufacturing goods but also includes the provision of services such as education, transportation, banking, and healthcare. The four main factors of production are:

Land – natural resources such as soil, minerals, forests, and water.

Labor – Human effort, both physical and mental.

Capital – Man-made tools, machinery, and equipment used in production.

Entrepreneurship – The ability to organize other factors and take risks.

The goal of production is to maximize output from given inputs or minimize cost for a given output level.

The production function shows the maximum output that can be produced from given quantities of inputs, assuming a given level of technology. It can be written as:

$$Q = f(L, K).$$

Where:

Q = Quantity of output,

L = Labor and

K = Capital.

The production function represents a purely technical relationship between inputs and output. It assumes technology remains constant, inputs are efficiently used, there is no waste and inputs are divisible and measurable. When technology improves, the production function shifts upward because more output can be produced with the same inputs.

In microeconomic theory, understanding the distinction between the short run and the long run is essential for analyzing how firms make production decisions. These concepts do not refer to specific calendar periods such as days or years; rather, they describe the flexibility a firm has in adjusting its factors of production. The short run is a period during which at least one factor of production remains fixed. In most cases, capital (K)—such as factory buildings, heavy machinery, and large equipment—is fixed because it cannot be easily or quickly adjusted. But, firms can vary certain inputs, especially labor (L), by hiring additional workers, increasing working hours, or reducing staff. As a result, short-run analysis focuses on how output changes when a variable input is combined with a fixed input. In contrast, the long run is a period during which all factors of production are variable. During this time, firms have complete flexibility to adjust plant size, install new machinery, adopt new technology, and change the scale of operations. There are no fixed factors in the long run, and firms can fully respond to

changes in demand, costs, and market conditions. The distinction between short run and long run is crucial because production behaviour differs significantly in each case.

Short Run

In the short run, at least one factor of production remains fixed. Typically, capital (K)—such as machinery, factory buildings, equipment, and technology—is considered fixed because it cannot be easily changed within a short period of time. On the other hand, labor (L) is treated as a variable factor since firms can adjust the number of workers more quickly by hiring or laying off employees. Under these conditions, the production function can be written as:

$$Q = f(L, \bar{K})$$

Where,

Q = Quantity of output,

L = Variable labor input and

$\bar{K}$ = Fixed capital.

This form of the production function shows that output depends on changes in labor while capital remains constant. Since capital is fixed, the only way to increase output in the short run is by increasing the amount of labor employed. As additional units of labor are combined with a fixed amount of capital, output does not increase in a simple linear manner. Instead, the relationship between labor and output follows a distinct pattern:

1. Initially, output increases at an increasing rate. When a small number of workers are employed, the fixed capital may be underutilized. As more workers are added, they can specialize in different tasks, coordinate better, and make more efficient use of machinery. This leads to better productivity and increasing returns.

2. After some point, output increases at a decreasing rate. As more workers are hired, the fixed capital becomes crowded. Each additional worker has less machinery or workspace available, reducing efficiency. Output still increases, but the rate of increase slows down.

3. Eventually, output may stop increasing or even decline. If too many workers are employed relative to the fixed capital, overcrowding and inefficiencies dominate. Additional workers may interfere with one another, and total output may decrease.

This changing relationship between labor and output gives rise to three important concepts in production theory namely Total Product, Marginal Product Labor and the Average Product of Labor.

Total Product (TP)

Total Product refers to the total quantity of output produced by a firm for a given quantity of labor, while holding other inputs—particularly capital—constant. In short-run production analysis, TP shows how total output changes as additional units of the variable factor (labor) are employed along with a fixed amount of capital. In other words, TP measures the overall productive performance of a firm at different levels of labor employment when plant size, machinery, and equipment remain unchanged.

We can explain total product with the help of a table. The table shows how total output changes as additional workers are employed while capital remains fixed. That is here the labour is the variable factor.

Table 12.1: Total Product

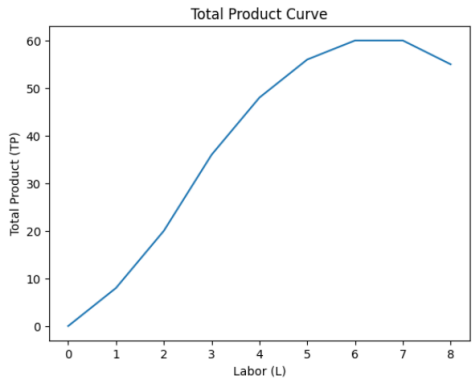
Labour	Total Product
0	0
1	8
2	20
3	36
4	48
5	56
6	60
7	60
8	55

From the table we can see that when labor = 0 i.e., when no workers are employed, total output is zero. Even though capital (machinery, building) is present, production cannot take place without labor. This highlights the importance of labor as an essential factor of production. From 0 to 3 workers we can see the Increasing Rate of Growth. Here output increases from 0 to 36 units. The increase in output becomes larger with each additional worker. During this stage, TP increases at an increasing rate. This happens because, fixed capital is initially

underutilized, workers can specialize in different tasks, division of labor improves efficiency and better coordination increases productivity. This phase reflects increasing returns due to improved utilization of the fixed factor. From 3 to 6 workers we can see a Decreasing Rate of Growth. Here output continues to increase, but at a slower rate that is TP rises from 36 to 60 units. Although output is still rising, each additional worker contributes less to total output compared to earlier workers. Total output increasing at a decreasing rate because fixed capital becomes more fully utilized, workspace becomes crowded, workers begin to compete for limited machinery and coordination becomes more difficult. That is this phase shows diminishing returns, where TP increases but at a decreasing rate. After 6 workers we can see Maximum and Decline. At 7 workers, TP remains at 60 and at 8 workers, TP falls to 55. At this stage the fixed capital is overutilized, overcrowding occurs, workers interfere with each other and efficiency declines significantly. Here, the firm enters the stage of negative returns. Additional labor reduces total output instead of increasing it.

If we plot these table into graph we get the total product curve.

Figure 12.1: Total Product Curve



If we plot labor on the horizontal axis and total product on the vertical axis, the TP curve will initially rise at an increasing rate (convex shape) then rise at a decreasing rate (concave shape). After that reaches a maximum and then eventually decline. The slope of the TP curve at any point represents the Marginal Product of Labor.

Marginal Product of Labor (MPL)

Marginal Product of Labor refers to the additional output produced by employing one more unit of labor while keeping all other factors of production—especially capital—constant. Since this concept is analyzed in the short run, capital remains fixed and only labor is variable. Therefore, MPL measures how total output changes when the quantity of labor alone is

increased. In simpler terms, MPL shows how much extra production a firm obtains by hiring one additional worker. It reflects the productivity of the last worker employed in the production process. The general formula for Marginal Product of Labor is:

$$MPL = \Delta TP / \Delta L$$

Where,

ΔTP = Change in Total Product and

ΔL = Change in Labor.

Since labor in our example increases by one unit at a time, the formula simplifies to:

$$MPL = TP_n - TP_{n-1}$$

This means MPL is the difference between the current level of total product and the previous level.

Table 12.2: MPL Calculation

Labor	TP	MPL
1	8	8
2	20	12
3	36	16
4	48	12
5	56	8
6	60	4
7	60	0
8	55	-5

When the workers are increased from 1 to 3 we can see an Increasing Marginal Returns. That is MPL rises from 8 to 12 to 16 and each additional worker contributes more output than the previous one. This happens because of better utilization of fixed capital, specialization and division of labor, improved coordination among workers and underutilized machinery becomes fully used. This phase reflects increasing returns due to improved efficiency. When increase workers from 4 to 6 we can see Diminishing Marginal Returns. That is MPL begins to fall from 16 to 12 then to 8 and then to 4. But the total output is still increasing, but at a decreasing rate.

This happens because fixed capital becomes crowded, workers share limited machinery, coordination problems arise and each additional worker has less capital to work with. This phase demonstrates the Law of Diminishing Marginal Returns. Further increment from 6 to 7 leads to Zero Marginal Product. MPL becomes zero. But we can see that total product reaches its maximum (60 units). The 7th worker adds nothing to output. At this point, all fixed resources are fully utilized and additional labor does not increase output. When we increase labour from 7 to 8 there occur Negative Marginal Product. MPL becomes negative that is (−5). Here we can see total product falls from 60 to 55. The 8th worker actually reduces total output, overcrowding causes inefficiency, workers interfere with each other and production becomes disorganized. This stage reflects negative returns. That is, marginal product is the slope of the total product curve.

When TP rises at an increasing rate

MPL is rising.

When TP rises at a decreasing rate

MPL is falling but positive.

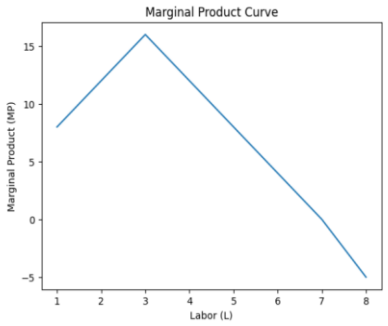
When TP is maximum

MPL = 0.

When TP declines

MPL is negative.

Figure 12.2: MPL Curve



From the graph we can see that MPL increases initially. Then it decreases, after that it becomes zero and later negative. Thus, the MPL curve first rises, reaches a maximum, then declines, crosses the horizontal axis, and becomes negative.

Average Product of Labor (APL)

Average Product of Labor refers to the output produced per unit of labor employed in the production process. It measures the average productivity of each worker by showing how much total output is generated, on average, by one unit of labor. APL is obtained by dividing total product (TP) by the quantity of labor (L) used in production. In simple terms, APL tells us how much output each worker produces on average. the formula for calculating

$$APL = TP/L$$

Where,

TP = Total Product and

L = Units of Labor.

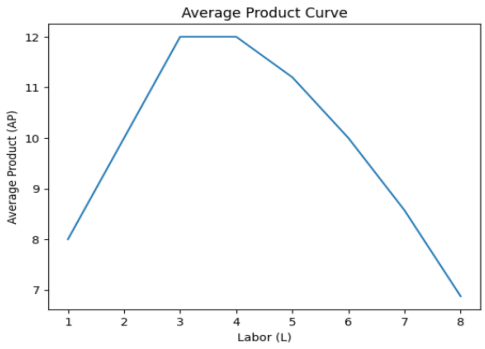
APL reflects the overall efficiency of labor in the production process. It indicates how effectively workers are utilizing the available fixed factors such as machinery, land, and equipment. When APL increases, it suggests that labor is being used more efficiently. When APL decreases, it indicates that labor is becoming less productive on average, often due to overcrowding of fixed resources. It is important to distinguish APL from Marginal Product of Labor (MPL). While APL measures the average output per worker, MPL measures the additional output produced by employing one more worker. Thus, APL focuses on overall productivity, whereas MPL focuses on the contribution of the last worker hired. In short, Average Product of Labor is a key concept in production theory because it helps firms evaluate labor efficiency, determine the optimal use of workers, and understand the stage of production in the short run.

Table 12.3: APL Calculation

Labor	TP	APL
1	8	8
2	20	10
3	36	12
4	48	12
5	56	11.2
6	60	10

From the table we can see that when only one worker is employed, total output is 8 units. Since there is only one worker, the entire output belongs to that worker. So, the average product is 8 units. When 2 workers are employed APL = 20 / 2 = 10. With two workers, total output increases to 20 units. On average, each worker now produces 10 units. The increase from 8 to 10 shows better utilization of fixed resources and possible cooperation between workers. When 3 workers are employed APL = 36 / 3 = 12. When three workers are employed, total output rises to 36 units. Each worker produces 12 units on average. This increase indicates improved efficiency due to specialization and division of labor. When 4 workers are employed APL = 48 / 4 = 12. With four workers, total output becomes 48 units. The average product remains 12 units. This suggests that productivity has reached its maximum level and is no longer increasing. When 5 workers are employed APL = 56 / 5 = 11.2. When a fifth worker is added, total output rises to 56 units. Though, the average product falls to 11.2 units. This means the additional worker contributes less than the existing average, causing the average to decline. When 6 workers are employed APL When 2 workers are employed = 60 / 6 = 10. With six workers, total output increases slightly to 60 units. The average product decreases further to 10 units. This indicates diminishing returns — additional workers are contributing less to total output due to limited fixed resources. As an overall we can say that APL increases from 8 to 12 (from 1 to 3 workers), APL reaches its maximum at 3–4 workers (12 units) and APL starts decreasing after the 4th worker. This pattern clearly shows the operation of the Law of Diminishing Marginal Returns, where average productivity first rises due to efficiency gains and then falls due to overcrowding of fixed factors.

Figure 12.3: APL Curve



From the figure, we can observe that APL initially increases from 8 to 10 to 12. This happens due to better utilization of fixed factors and specialization. Then APL reaches its maximum at 3–4 workers that is 12 units. Finally, APL starts decreasing after the 4th worker that is from 12 to 11.2 to 10. This occurs because of overcrowding of fixed factors and diminishing returns.

Thus, these calculations help identify the most efficient level of labor employment in the short run.

In the short run, Total Product, Marginal Product of Labor and Average Product of Labor together explain the behaviour of production when one factor (labor) is variable and others are fixed. Total Product initially increases at an increasing rate, then at a decreasing rate, reaches a maximum, and may eventually decline. Marginal Product of Labor first rises due to better utilization and specialization, then falls due to diminishing returns, becomes zero when TP is maximum, and turns negative afterward. Average Product of Labor rises when MPL is greater than APL, reaches its maximum when MPL equals APL, and declines when MPL is less than APL. Overall, these concepts demonstrate the Law of Diminishing Marginal Returns, showing that after a certain level of labor employment, additional workers contribute less to output due to fixed resources.

Multiple Choice Questions

1. In the context of short-run production, which factor of production is typically considered fixed?

A. Labor

B. Capital

C. Land

D. Entrepreneurship.

Answer: B. Capital.

2. What is the Average Product of Labor (APL) when total output is 36 units and 3 workers are employed?

A. 10 units

B. 12 units

C. 8 units

D. 11.2 units

Answer: B. 12 units.

3. The Law of Diminishing Marginal Returns indicates what effect on Average Product of Labor (APL) after a certain point in hiring workers?

A. APL continues to increase indefinitely.

B. APL reaches a maximum and then starts to decrease.

C. APL remains constant at all levels of employment.

D. APL is always lower than Marginal Product of Labor (MPL).

Answer: B. APL reaches a maximum and then starts to decrease.

4. If the Marginal Product of Labor (MPL) is greater than the Average Product of Labor (APL), what happens to the APL?

A. APL decreases.

B. APL remains the same.

C. APL increases.

D. APL becomes zero.

Answer: C. APL increases.

5. Which statement best describes the production function in microeconomics?

A. It shows the relationship between input prices and output levels.

B. It illustrates the maximum output produced from given quantities of inputs.

C. It defines how firms allocate fixed resources.

D. It measures the demand for goods and services in the market.

Answer: B. It illustrates the maximum output produced from given quantities of inputs.

Law of Diminishing Marginal Returns

Production is one of the central activities in economics. It refers to the process of transforming inputs such as land, labor, capital, and entrepreneurship into goods and services that satisfy human wants. Since resources are scarce and human wants are unlimited, efficient production becomes essential for economic success. Firms must carefully decide how much of each input to use in order to maximize output and profits. In microeconomic theory, the short-run production function plays a vital role in understanding how output responds to changes in inputs. In the short run, at least one factor of production remains fixed while others are variable. Typically, capital (machinery, land, factory space) is fixed, and labor is considered the variable input. The relationship between the variable input and output gives rise to three important concepts TP, MPL and APL. These concepts lead to one of the most fundamental principles the Law of Diminishing Marginal Returns, first systematically discussed by classical economists such as David Ricardo and Thomas Robert Malthus. This law explains why additional units of a variable factor eventually produce smaller increments in output when other factors are fixed.

The Law of Diminishing Marginal Returns states that as more units of a variable input are added to a fixed input, the marginal product of the variable input eventually declines, holding technology constant. The Law of Diminishing Marginal Returns states that when additional units of a variable input, such as labor, are added to a fixed input like capital or land, the marginal product of that variable input will eventually begin to decline, assuming that technology remains constant. In the short run, certain factors of production cannot be increased, so as more workers are employed with the same amount of machinery, equipment, or land, each additional worker has less fixed capital to work with. Initially, productivity may rise due to better utilization of resources and specialization, but after a certain point, overcrowding and resource limitations reduce efficiency. As a result, each new worker contributes less additional output than the previous one. This gradual decrease in the extra output produced by successive workers is the essence of the Law of Diminishing Marginal Returns.

To understand the law clearly, consider the short run, where at least one factor of production remains fixed. In most economic analysis, capital (such as machinery, tools, factory space, or land) is treated as the fixed factor, while labor is considered the variable factor. This means that output changes mainly because of changes in labor, while capital remains constant. The law explains how output behaves when additional units of labor are combined with a fixed

quantity of capital. The process can be understood in three distinct phases: Initial Phase, Intermediate Phase, and Final Phase.

1. **Initial Phase**

Initial Phase is also known as Increasing Returns Phase. This occurs at the early stage of short-run production when only a small number of workers are employed with a fixed amount of capital. At this stage, the firm has not yet fully utilized its fixed resources such as machinery, equipment, tools, factory space, or land. Because capital is fixed in the short run, the early addition of labor allows the firm to make better use of these underutilized resources. At the beginning of production only a few workers are employed, fixed capital such as machines, tools, and workspace remain partly unused, there is excess production capacity, workers have sufficient space and access to equipment and supervision and coordination are easy due to the smaller workforce. Under these favourable conditions, adding more workers leads to a more than proportionate increase in total output. This means output increases at an increasing rate. The main reasons behind this improvement are:

1. Better Utilization of Fixed Capital: When very few workers are employed, machines and equipment cannot be fully operated. For example, a factory may have several machines but too few workers to operate them simultaneously. As additional workers are hired idle machines are brought into operation, production facilities are fully used and the fixed factor is utilized more efficiently. Since capital remains constant while labor increases, the additional workers allow the firm to extract maximum benefit from existing resources. This improves overall productivity.

2. Division of Labor: With the increase in the number of workers, tasks can be divided into smaller and more specialized activities. Instead of one worker performing multiple tasks inefficiently, different workers can focus on specific tasks. Division of labor results in faster completion of tasks, reduced time wastage, higher skill development and improved production speed. When work is divided systematically, each worker becomes more efficient in their assigned role, which significantly increases output.

3. Specialization: Specialization is closely related to division of labor. As workers repeatedly perform specific tasks, they gain expertise and improve their efficiency. Specialization leads

to greater accuracy, reduced errors, increased speed and better-quality output. Because each worker concentrates on a particular activity, productivity improves and marginal product rises.

4.Improved Cooperation and Teamwork: In the early stages, a moderate increase in workforce enhances cooperation and teamwork. Workers can assist one another, share responsibilities, and coordinate tasks efficiently. Additionally, communication is easier in smaller groups, supervision is more effective, management control is stronger and coordination costs remain low. These factors create a productive environment that supports rapid growth in output.

5. Economies of Scale Within the Short Run: Even within the short run, small internal efficiencies may arise when labor increases initially. These efficiencies contribute to increasing returns, as resources are better organized and utilized. Because of all these favourable conditions total product increases rapidly, output rises at an increasing rate, marginal product rises, meaning each additional worker contributes more than the previous one, the total product (TP) curve becomes convex in this region and the marginal product (MP) curve slopes upward. This phase highlights the importance of efficient organization, specialization, and teamwork in improving productivity. But, this phase does not continue indefinitely. Since capital remains fixed, there comes a point where adding more labor no longer leads to rising marginal returns. Eventually, production moves into the second stage where diminishing returns begin to operate.

2. Intermediate Phase

After a certain number of workers have been employed, the production process moves from the initial increasing returns phase into the Intermediate Phase, commonly known as the Diminishing Returns Phase. This stage represents the central operation of the Law of Diminishing Marginal Returns and is the most important phase in short-run production. At this stage fixed capital becomes relatively scarce compared to labor, each worker has access to less machinery, tools, or workspace, workers begin sharing equipment, mild congestion starts appearing in the production area and supervision becomes slightly more complex. Although total output continues to increase during this phase, the rate at which it increases begins to slow down. This means output is still rising, but not as rapidly as before. The main reason is the limited quantity of the fixed factor. In the short run, capital such as machines, factory buildings, or land cannot be increased. As more workers are added the labor-to-capital ratio increases, each additional worker has less capital to work with and the effectiveness of additional labor

gradually declines. In simple terms, when too many workers are combined with a fixed number of machines, each worker cannot work as efficiently as before. Thus, while total production rises, the extra output produced by each new worker becomes smaller.

In the Intermediate Phase Total Product (TP) still increases, but at a decreasing rate, The TP curve becomes concave, Marginal Product (MP) starts falling, although it remains positive and each additional worker adds output, but less than the previous worker. This is the true operation of the Law of Diminishing Marginal Returns. It is important to understand that diminishing returns do not mean negative returns. Total output is still increasing. The only change is that the additional contribution of each new worker becomes smaller. In this rational stage two the firm is fully utilizing its fixed resources, output is still increasing, additional labor still adds value and production remains profitable. A rational producer will operate in this stage because employing additional workers still increases total output, even though at a decreasing rate.

Final Phase

When labor continues to be added beyond the optimal level of employment, production enters the Final Phase, also known as the Negative Returns Phase. This stage represents the extreme consequence of overusing the variable factor while the fixed factor remains constant in the short run. At this stage severe overcrowding occurs in the workplace, Workers begin to interfere with each other's tasks, machinery becomes excessively overused, coordination and communication become difficult and supervision loses efficiency due to too many workers. Since capital such as machines, tools, and workspace cannot be increased in the short run, the excessive number of workers creates imbalance in the production process. The fundamental reason is the extreme imbalance between labor and fixed capital. When too many workers are combined with limited machines some workers remain idle waiting for equipment, work overlaps and duplication increase, errors and accidents become more frequent, time is wasted in adjustment and coordination and productivity per worker falls sharply. In the Final Phase marginal product becomes zero at the beginning of this stage. If more workers are added beyond this point, marginal product becomes negative. Total product reaches its maximum when MP is zero. After this point, total product starts to decline because negative marginal product subtracts from total output. A rational producer will never operate in this stage because Additional labor reduces total output, marginal product is negative and profit decreases. Therefore, the firm will stop employing workers before reaching this stage.

The entire process demonstrates that:

-Initially, adding labor increases productivity due to better utilization and specialization.

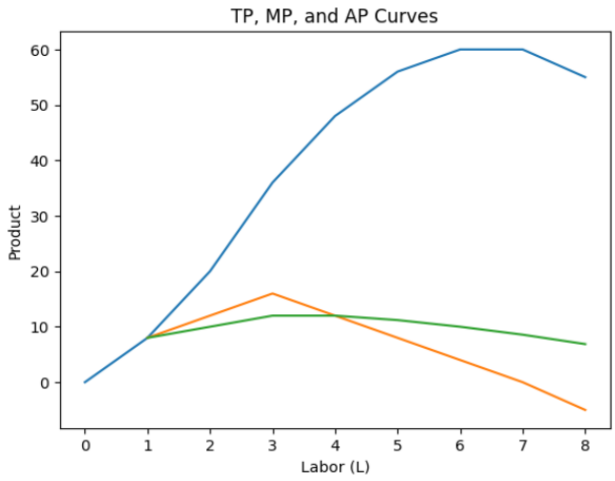
-Eventually, fixed capital becomes insufficient relative to labor.

-Additional workers contribute progressively less output.

- Beyond a certain point, extra workers may reduce total output.

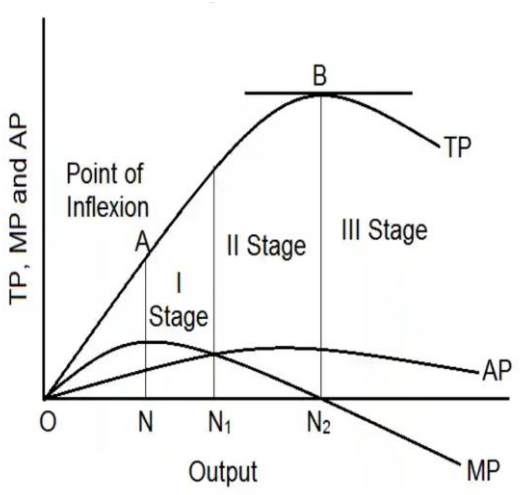
This gradual decline in the additional output produced by successive units of labor is the core idea behind the Law of Diminishing Marginal Returns. Thus, the law highlights the importance of maintaining an optimal balance between labor and capital in order to ensure efficient production and maximum profitability.

Figure 12.4: TP, MP and AP Curves



From the figure we can see that when no workers are employed total output is zero. Even though capital (machinery, building) is present, production cannot take place without labor. From 0 to 3 workers increasing rate of growth, 3 to 6 workers decreasing rate of growth and 6 to 8 workers decreasing rate of growth can be seen. This can be generalised with the help of a figure.

Figure 12.5: TP, MP and AP Curves



In the figure output is shown in the X axis and Y axis shows TP, MP and AP. From 0 to N_1 first stage or increasing returns, from N_1 to N_2 second stage or diminishing returns and beyond N_2 third stage or negative returns occurred. These stages help to explain how output changes as more units of a variable input, such as labor, are combined with a fixed input like capital.

Stage I: Also known as the Increasing Returns Stage, is characterized by total product increasing at an increasing rate. In this phase, both marginal product and average product rise. This stage ends when average product reaches its maximum point. The reason for increasing productivity is that fixed resources are initially underutilized, and the addition of more workers allows better use of machinery and workspace. Specialization and division of labor improve efficiency, enabling workers to complement one another and produce more output. Both MPL and APL rise because each additional worker contributes more than the previous one. But, it is irrational for a firm to stop production in this stage since average productivity is still rising and fixed resources are not fully utilized. Stopping here would mean underusing productive capacity and not achieving maximum efficiency.

Stage II: Also known as the Diminishing Returns Stage, is the most important and rational stage of production. In this phase, total product continues to increase but at a decreasing rate. Marginal product begins to decline, although it remains positive, and average product also starts falling. This stage ends when marginal product becomes zero. Here, the fixed factor is fully utilized, and while each additional worker still adds to total output, the extra contribution becomes smaller. This is where the Law of Diminishing Marginal Returns operates clearly. The decline in marginal product occurs because more labor is combined with a fixed amount of capital, reducing efficiency per worker. Firms operate in this stage because marginal product

is positive, total output is increasing, and profits can still be earned. It represents optimal utilization of both fixed and variable factors.

Stage III: Also known as the Negative Returns Stage, occurs when labor is increased beyond the optimal level. In this stage, total product begins to decline and marginal product becomes negative. Overcrowding becomes severe, workers interfere with each other, machines are overused, and coordination problems arise. Instead of increasing output, additional workers reduce total production due to inefficiency and congestion. No rational firm will operate in this stage because hiring more workers increases costs while reducing output and profits. Therefore, firms avoid this stage and aim to operate within Stage II, where production remains efficient and profitable.

Table 12.4: Comparison of TP, APL and MPL

Stage	TP	MPL	APL	Nature
I	Rising fast	Rising	Rising	Irrational
II	Rising slowly	Falling (+)	Falling	Rational
III	Declining	Negative	Falling	Irrational

Stage I:

1. Total Product rises rapidly because production increases at an increasing rate.
2. Marginal Product is rising, which means each additional worker contributes more output than the previous one.
3. Average Product is also rising, showing that productivity per worker is improving.

This happens because fixed resources such as machines and land are underutilized, and additional workers help use them more efficiently through specialization and division of labor. This stage is considered irrational for a firm to stop production because resources are not fully utilized and productivity is still increasing. The firm can increase output further by employing more workers.

Stage II:

1. Total Product continues to rise but at a slower rate.
2. Marginal Product begins to fall, though it remains positive.

3. Average Product also declines because marginal product is now below average product.

This stage reflects the operation of the Law of Diminishing Marginal Returns, where the fixed factor becomes fully utilized and each extra worker has less capital to work with. Despite diminishing returns, this stage is considered rational because total output is still increasing, marginal product is positive, and profits can still be earned. Firms aim to operate within this stage since it represents optimal utilization of resources.

Stage III:

1. Total Product starts to decline.
2. Marginal Product becomes negative, meaning additional workers actually reduce total output.
3. Average Product continues to fall due to severe overcrowding and inefficiency.

Too many workers relative to fixed capital cause congestion, coordination problems, and reduced productivity. This stage is **i**rrational because hiring additional labor increases costs while reducing output and profits. No rational firm would choose to operate here.

The Law of Diminishing Marginal Returns is a cornerstone of short-run production theory. It explains how output behaves when additional units of a variable input are added to a fixed input. Initially, productivity increases due to specialization and better utilization of fixed factors. But, as more labor is added, fixed capital becomes insufficient, leading to declining marginal productivity. Eventually, excessive labor results in negative returns. Firms will operate in the second stage where output increasing at a decreasing stage. That is this theory helps to achieve efficient resource allocation and maximum profit.

Multiple Choice Questions

1. What does the Law of Diminishing Marginal Returns state?

A. As more units of a variable input are added to a fixed input, the total product will continue to increase indefinitely.

B. As more units of a variable input are added to a fixed input, the marginal product of the variable input eventually declines.

C. Increasing the fixed input will lead to higher marginal returns of the variable input.

D. The marginal product of labor remains constant regardless of the number of workers employed.

Answer: B. As more units of a variable input are added to a fixed input, the marginal product of the variable input eventually declines.

2. In the context of the Law of Diminishing Marginal Returns, which factor is typically considered fixed in the short run?

A. Labor

B. Capital

C. Technology

D. Raw materials

Answer: B. Capital

3. During which phase does the Law of Diminishing Marginal Returns begin to take effect?

A. When no workers are employed

B. When the maximum number of fixed inputs is utilized

C. After a certain point of adding workers to fixed inputs, leading to decreased marginal productivity

D. When variable inputs exceed fixed inputs

Answer: C. After a certain point of adding workers to fixed inputs, leading to decreased marginal productivity.

4. Which of the following scenarios best illustrates the Law of Diminishing Marginal Returns?

A. A factory increases production by investing in more machines.

B. A restaurant hires too many cooks, resulting in confusion and reduced meal preparation efficiency.

C. A farmer expands his land to grow more crops, increasing overall yield.

D. A tech company hires additional engineers, significantly boosting innovation.

Answer: B. A restaurant hires too many cooks, resulting in confusion and reduced meal preparation efficiency.

5. What is the primary outcome when more units of a variable input are added to a fixed input according to the Law of Diminishing Marginal Returns?

A. Total output decreases immediately.

B. Marginal product increases with each additional unit of input.

C. Marginal product eventually declines after an initial increase.

D. Average product remains unaffected regardless of input changes.

Answer: C. Marginal product eventually declines after an initial increase.